

Deep learning driven Pneumonia detection using Xray images

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Abstract—Major deaths (55% or 1.5 million) of LRI(Lower Respiratory Infections) are caused due to *Pneumococcal pneumonia*. LRI is the 5th highest cause of demise as mentioned by the Global Burden of Disease Study 2015. Health organization suffers from a lack of Physicians. A Physician handles so many cases which leads to misdiagnoses. To resolve all these issues we came up with an easy-to-handle, and implemented in health organizations, the CNN model. The main aim of this model is to help the healthcare professionals by providing them with a tool which will help them detect *Pneumococcal pneumonia*. With the help of this tool, the work strain of a doctor can be reduced. This model is developed without any pre-trained model and for better accuracy, an immense dataset is used. All the X-ray images were preprocessed before being introduced to the neural network. Different layers have been used to make the model more precise and make it adaptable for X-ray images under different conditions. The aim is to combine deep learning concepts with healthcare to make a positive and meaningful impact in the healthcare industry.

I. LITERATURE REVIEW

The increase of applied Machine Learning and Deep Learning in various fields brings attention of the Tech Critics on the application of both in the wide field of Medical X-ray diagnosis and Pneumonia detection. A paper by Varshini et al.,[1], targets the applications of convolution-based Deep Learning neural network architecture studied from previous researches but improved and made more intelligent. The aim is not to replace the doctor's diagnosis but to make them more accurate and faster.

A few studies done by; Bagaria et al.,[2] explain the basic per-processing methods, and segmentation methods, and classify the X-ray image data set with an error back propagation neural network. Their approach involves the use of a relatively small data set of 530 images, augmented through techniques like rotation, and flipping to enhance the training period of the model. A work on detecting Pneumonia done by Rahman et al.,[3] uses pre-trained models such as Alex-net, ResNet18, DenseNet201, and Squeeze-net. Large pre-trained models with 100000+ weights are complex to understand, their architecture is

complex and the model is more biased towards what it was trained for originally.

In a previous research conducted by Hussain et al.[4], a new approach for categorization of Chest X-rays using a machine learning-based approach was found. However, the algorithm that was developed following their approach was trained on a very short public data set.

An interesting approach with multi-class medical classification done by Tiwari et al.[5] has given an overview of how CNN can be optimized for good accuracy and complex detection in medical scans. All in all, research conducted by Yang et al. [6] is an in-depth and complete study on how object detection models can be used by the help of Semantic Segmentation in the field of medical diagnosis. Their in-depth study on how the CNN constructed algorithm can be used as a backbone for pre-trained models for specific model training can be used to increase the accuracy and precision of the model.

The model developed in this research paper deals with the limitation of a smaller data set used for the training of a X-ray pneumonia detection model by using a 3600+ image data set. We have increased the number of Max Pooling layers and Convolutional layers to get more precise details for Pneumonia detection. There are no pre-trained models used in this research. All of the research conducted in this paper results in an intelligent and robust model for X-ray detection using a larger data set for a wider and more complex detection in the medical scans.

II. INTRODUCTION

Chest X-ray creates images of all organs and pipes that are present in the chest area. It also shows the air and fluid surrounding the lungs and detects any fluid present inside the lungs.

The study of classification and detection of major diseases from X-rays has become a challenging task. In this solution we consider Pneumonia. Pneumonia can be easily identified using various chest scans. X-rays are used to diagnose many lung diseases because they are fast, easy, and inexpensive as compared to other scans.

Pneumonia is (ranked 4th) a leading cause of demise across all ages, and in 13 seconds 1 person dies because of

pneumonia. Till now more than 808,000 children died under the age of 5 in 2017, which is at least 15% of all deaths of preschoolers. It causes the air sacs to become inflamed, and due to this, breathing becomes challenging which leads to the swelling of the lungs and fluid accumulation in the lungs.

- This study shows that using more and relevant features from a deep learning model can significantly increase performance. In this work, a dataset of infected chest X-ray of Pneumonia is used to train and extract relevant features using a deep learning model. Here is the list of what was contributed.
- We trained and tested the CNN model on a data set of (5216, 624,16) 5856 images which fully interpret the disease by extracting their features and properties from images.
- This model was achieved by following three steps:
 1. Training a CNN model;
 2. Comparing the pre-trained model's accuracy and losses;
 3. To avoid over-fitting , optimal batch size, added a dropout layer, introduction to L2 regularization, and early stopping in the model;

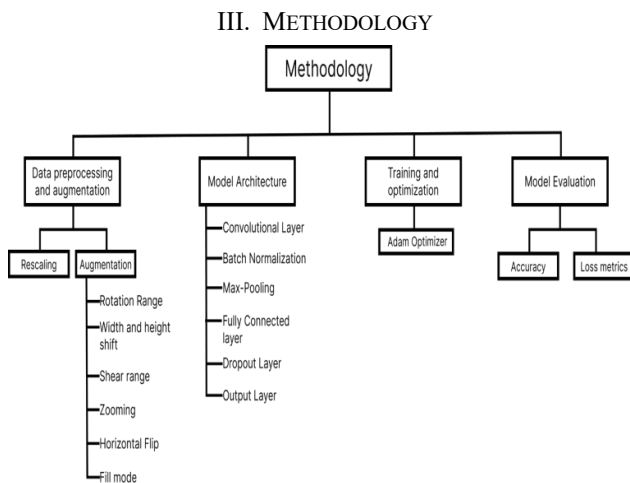


Fig 3.1 Flow chart of the entire workflow of the research.

This model processes the Chest X-ray images and detects pneumonia with the help of Deep learning techniques. The data-set which is used to make this model is a comprehensive data set of X-ray images, these images are classified into two classes: **pneumonia-affected chest X-rays and normal chest X-rays** (Kermany, Daniel,2018).[7] A study done by Kaori Togashi et al.[8] was a beneficial guidance as it explains the potential of CNN to improve a radiologist's performance and patient care. Their research shows the efficiency of CNN in medical images due to its ability to capture spatial hierarchies and

handling larger datasets. The system consists of 4 modules. A deeper dive into the modules:

1. Data preprocessing and augmentation:

Preprocessing allows to remove any distortion or noise in the data set. Preprocessed data reduces the training time of the model and makes it more adaptable for real-world scenarios. Here, the following processes are done in data preprocessing:

A. Re-scaling:- An image is in the form of pixels and these pixel values range from 0 - 255 where 0 is for black and 255 for white and the values between them represent different colors. The method used to normalize the pixel value is simple, it works by dividing each pixel value by 255 (the standard value used in deep learning models) which converts the pixel into a value ranging between 0 to 1. For this model, the pixel values are converted to [0,1] because normalized values are easier to compute and improve the numerical stability during training. The formula is :-

$$\text{Re-scaled value} = \frac{\text{Original Value}}{255} \quad (1)$$

Equation 1 - Formula for gray-scaling / toning down the color scale of a given image.

Example- A pixel value of 163 becomes 0.639 (approximately)

$$\text{Re-scaled value} = 163 / 255$$

B. Augmentation:-Image data augmentation is a way of creating more and new data set by performing a set of operations on the original data. It was done while keeping in mind the changes caused by data augmentation on a CNN model's performance by Raina et al[9]. Through this, model adapts to process X-Ray images in all different angles, orientations, size, etc.

Following techniques have been used to apply image augmentation:-

1) **Rotation Range-** It randomly rotates the images to a specified degree. The rotation range is set to 20 degrees for this particular model which means the images will be rotated in the X and Y axis up to an angle of 20 degree. This allows the model to work on images with different orientations, and mathematical explanations: This rotation is performed with the help of a Rotation matrix(a matrix used to perform rotation in Euclidean space).

Representation:

$$R = \begin{bmatrix} \cos\Theta & -\sin\Theta \\ \sin\Theta & \cos\Theta \end{bmatrix} \quad (2)$$

Equation 2- Representation of the rotation matrix.

According to this representation, this matrix rotates the points or in this case the pixel values of a 2-dimensional image in the xy plane in counter-clockwise direction through the given angle (here, 20 degrees) about the origin or the center of the image.

2) Length and Breadth shifting -The function is to randomly shift the image's height and width by some percentage of the total size. The range is set to 20% for this model.

3) Shear range- Shearing means tilting the object in the image; The parameter of shear range specifies the percentage up to which the image will be distorted in a particular direction. For example- Shearing in X-direction: horizontal shearing

$$\text{Matrix: } \begin{bmatrix} 1 & 0 & 0 \\ Sh_x & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

Equation 3 - Representation of the Shearing matrix in X direction

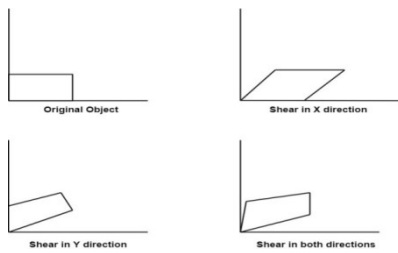


Fig. 3.2 Representation of shearing of an image in different directions.

4) Zooming- This simply enlarges or shrinks the image.

5) Horizontal flip- Horizontal flip is applied to make the model adaptable for X-ray in its mirror image form as well.

2. Model Architecture:

This model works on the concept of Convolutional Neural Network (CNN) architecture, it is a deep learning algorithm which is made up of multiple layers and this helps this algorithm to deal with large datasets. A research done by Hussain et al [10] has shown the potential of CNNs in medical image classification. The layers used in the bone fracture detection model are as follows:-

1. Convolutional layer: These layers detect features like edges, textures, and patterns by applying Convolutional operations using filters. This is the mathematical formula used for Convolutional operation:

$$(I * F)(i, j) = \sum_{m=0}^{M-1} \cdot \sum_{n=0}^{N-1} \cdot (i + m, j + n) \cdot F(m, n) \quad (4)$$

Equation 4- Here, I = input image; F = filter ; i,j = coordinates of the output feature map, M x N = size of the filter

In this model, the Convolution layers are used with the increased filter size of (32, 64, 128, 256, and 512).

2. Batch Normalization: This technique is used for enhancing the training speed and stability of the neural network's. This basically uses the output from a layer and adjusts its value such a way that the mean comes out to be 0 and the standard deviation is 1.

A research work by Ogundokun et al [11] shows that when batch normalization is combined with Adam optimizer, it significantly enhances the performance of CNN models.

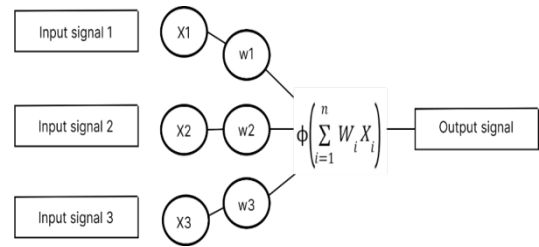


Fig 3.3 Representation of the neurons in a basic neural network.

3. Max-Pooling layers: This downscale the spatial dimensions and reduces the parameter count in the network. The given image is divided into small regions and then the maximum value is computed, this value forms the output of that region.

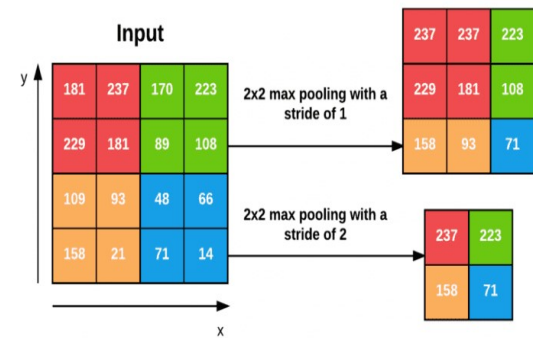


Fig 3.4 Representation of a 2x2 Max-Pooling layer with a stride of (1,2).

4. Fully Connected Layer: These layers predict the input using the features learned before. In this, a neuron in a layer is connected to every neuron in the next layer and every input is multiplied by a weight then a bias model is added. In this CNN network, the layer is 256 neurons, and the ReLU activation function is used.

Mathematically, it looks like this for one neuron:

$$y = \sigma(\sum (x_i \cdot w_i) + b) \quad (5)$$

Equation 5 - X_i is a result from the previous layer, W_i is the corresponding weight, b is the bias term, σ is the activation function (like ReLU or sigmoid), y is the output of the neuron.

ReLU function- This function is used to introduce a sense of non-linearity which helps the architecture to learn more complex patterns. Mathematically,

$$f(x) = \max(0, x) \quad (6)$$

Equation 6 - Representation of the ReLU activation function

- 1.If the input x is **+ve**, then the output is x.
2. If the input x is **-ve or 0**, the output is **0**.

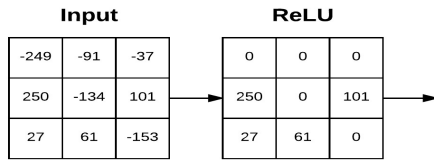


Fig 3.5-This figure shows the input volume going through a ReLU activation, $\max(0, x)$.

The use of Relu function, rather than the softmax for image classification, in the CNNs was done while reviewing the literature written by Agarap et al [12].

5. Dropout: This technique is used to prevent over-fitting and this is done by dropping a fraction of neurons at random during the training period. A study done by Salehin et al[13], uses a dropout probability of 0.5, which helps balances the model between underfitting and overfitting in neural networks.

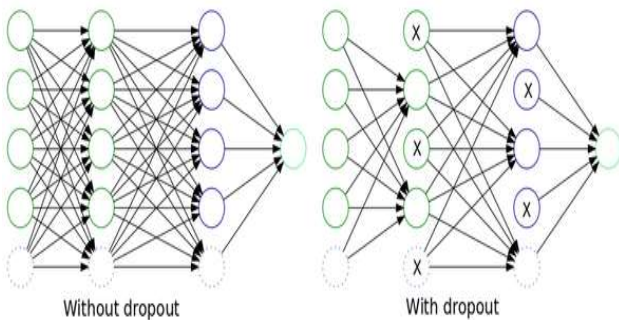


Fig 3.6 Visualization of Dropout layers in a Neural network

6. Output Layer- The last layer is the output layer of the model which is responsible for the final result. This layer is a single neuron layer with sigmoid activation.

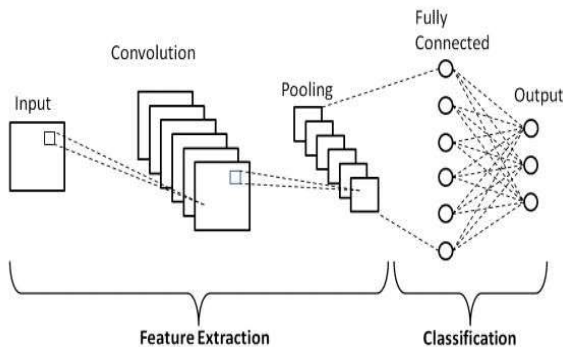


Fig 3.7 Here, is a complete diagram to understand the working of CNN architecture:

3. Training and Optimization:

Adam optimizer is taken to compile the model with the learning rate of 1×10^{-4} , and the callback. Early stopping is responsible to terminate the training if there is improvement for 10 consecutive epochs. The training is conducted for 20 iterations with a batch size equal to 16.

4. Model Evaluation:

After the model was trained it was tested on both training data and validating data. The accuracy and Loss metrics were used to determine the performance of the model. Building on the advanced study on how the evaluation metrics work done by Hicks et al[14], several visualisation techniques have been used for clear understanding, like Accuracy and Loss curves, confusion matrix heatmap, and ROC and PR curves.

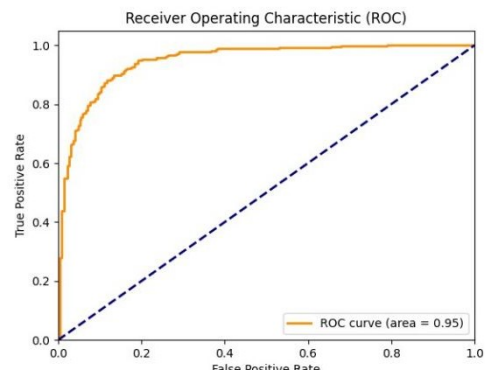


Fig 3.8 Representation ROC

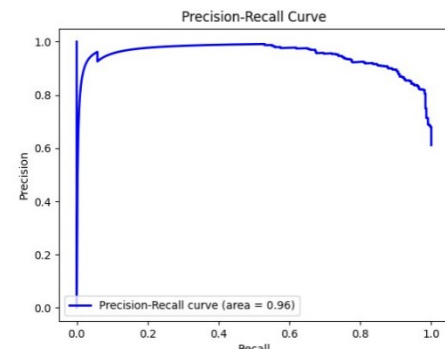


Fig 3.9 Precision recall curve

Score -

AUC	Pre AUC	Recall.	F1 Score
0.95	0.96	0.93	0.89

Fig 3.10 Different metric scores for the developed model

IV. RESULT AND DISCUSSION

After performing different precision and accuracy tests we arrive at a model with an accuracy of 87%, and after comparing this model's accuracy with the present models that are present in the market, this model achieves a higher accuracy while using a data-set that is much larger than the

previously used datasets. The F1 score of the model is 0.89, Recall being 0.932 and the precision is 0.8526. The True positive percentage of the testing data came out to be -

1. Normal True Positive - 89%
2. Correctly classified Pneumonia True Positive - 86%

A CNN model with 5 Convolutional layers and increasing filter sizes from 32 to 512. More of these layers help in capturing the image features. An L2 regularization in dense layers and dropout to mitigate over-fitting and generalize the model was used.

In Comparison with a previously done study on the detection of Pneumonia using chest X-rays by Hashmi et al [15] our model improves upon several limitations of transfer learning approaches, such as pre-trained model bias. It leverages a custom architecture tailored to chest X-ray images and pneumonia detection, potentially offering a more generalized solution without the biases of other domain-specific models.

Author	Technique	AUC
Enes AYAN et al [16]	VGG16 deep learning model	87
Xianghong Gu et al. [17]	Deep Convolutional neural network	82
A. A. Saraiva et al [18].	Neural network	94.5
This Model	CNN Model Architecture	95.89

Fig 3.11 Comparison from other models.

The above figure explains how our model compares to the others that have already tried to achieve this X-ray Pneumonia detection.

The learning rate with the Adam optimizer was kept at 1×10^{-4} , although image quality may vary and hence different accuracies are achieved.

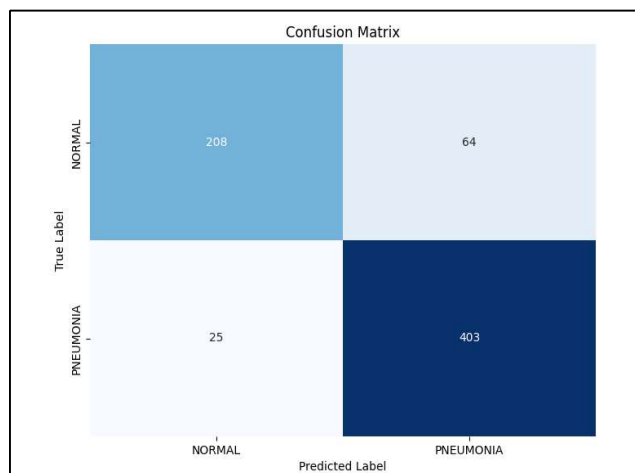


Fig 3.12 Confusion matrix.

The future prospect is that we aim to achieve with this model is to get more real-life data from various institutions and collect a more real-life and vast data set.

As the current model has shown significant accuracy with a robust data set, there are several key areas for further development and enhancement to optimize its real-world application. The future prospects of this project focus on improving both the technical and practical aspects of the model's deployment in medical diagnosis settings:

1. Expansion of the Data set: The most critical steps, is to gather more real-life data from a wider range of institutions, hospitals, and geographical locations. By acquiring data from diverse populations, the model is to be trained on a more comprehensive set of variations in X-ray imagery, including different age groups, regions, and lung conditions. This will enhance the model's generalization capabilities and adaptability to real-world scenarios.

2. Integration with Hospital Management Systems: The developed model can be integrated with an existing hospital management systems (HMS) to support radiologists in diagnosing pneumonia. This real-time integration could provide immediate feedback on X-ray scans, allowing doctors to make faster and more accurate decisions, especially in resource-limited settings like rural areas.

3. Mobile and Telemedicine Applications: Given the increasing use of mobile devices for health care, the model can be implemented in mobile health (mHealth) applications. By deploying the model on smartphones or tablets, healthcare professionals in remote or underdeveloped areas can gain access to a diagnostic tool for pneumonia without needing specialized equipment. This can also assist in telemedicine consultations, where radiological support is often scarce.

V. CONCLUSION

CNN model with 93.2%, 89.06%, 94.82%, 95.89% Recall, F1, AUC and PR AUC score. This reduces the chances of wrong diagnosis as 38.8% (n=73) of the hospitalized Pneumonia patients were diagnosed. Upper respiratory tract infection (32.8%, n=24) was the most common alternative diagnosis. Incorrectly diagnosed patients were relieved earlier (LOS= 3.5 ± 3.28 days vs. 7.7 ± 15.29 days, $p=0.03$) Even though there are other model with high accuracy but they are not in reach, need high computational power, or need prior knowledge. This CNN model also takes care of them.

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